

Question	Answer	Marks
2(a)	(work done =) force $\times$ distance <u>moved</u> in direction of force	B1
2(b)(i)	1. acceleration = gradient or $a = (v - u) / t$ or $a = \Delta v / t$	C1
	e.g. $a = 2.4 / 3.0$ $= 0.80 \text{ m s}^{-2}$	A1
	2. tension in cable = $(13.0 + 2.0) \times 10^3$	C1
	work done = $15 \times 10^3 \times (3.0 \times 2.4)$ $= 1.1 \times 10^5 \text{ J}$	A1
2(b)(ii)	power = Fv	C1
	$v = 2.0 \text{ (ms}^{-1}\text{)}$	C1
	input power = $(1.6 \times 10^4 \times 2.0) / 0.67$ $= 4.8 \times 10^4 \text{ W}$	A1
2(b)(iii)	work is done against friction so (increase in) GPE is less (than work done by motor) or energy is lost or transferred or converted to heat/thermal energy due to friction or resistance force or work is done lifting the cable so GPE is less	A1